

28	Two beams, P and Q, of light of the same wavelength fall upon the same metal surface causing photoemission of electrons. The photoelectric current produced by P is nine times that produced by Q.						
	Which of the following gives the ratio						
	$\frac{\text{wave amplitude of beam P}}{\text{wave amplitude of beam Q}} ?$						
A	0.11	B	0.33	C	3.0	D	9.0

Answer: C

photocurrent $\propto$ intensity

intensity $\propto$ (amplitude)²

$$\frac{\text{intensity of beam P}}{\text{intensity of beam Q}} = \frac{k \times (\text{wave amplitude of beam P})^2}{k \times (\text{wave amplitude of beam Q})^2} = \frac{9}{1}$$

$$\frac{\text{wave amplitude of beam P}}{\text{wave amplitude of beam Q}} = \sqrt{\frac{9}{1}} = 3$$

[Turn over

